Building a Multilingual Retrieval-Augmented Question-Answering System Using XRAG

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Abstract Access to reliable news can be difficult when important information is reported in a language a reader does not understand. This project describes the development of a multilingual retrieval-augmented generation system that processes news articles and answers questions in English, Spanish, or Arabic. The system was evaluated using 50 questions from XRAG: 25 Arabic and 25 Spanish. Four retrieval approaches were tested: document-level BM25, multilingual E5 over overlapping chunks, a BM25–E5 hybrid, and sentence-level BGE-M3. BM25 was strongest among the initial approaches, while BGE-M3 produced the best overall retrieval results. At Top-20, BGE-M3 represented both supporting articles for 40 of 50 questions and achieved 0.88 average support-article recall. In a preliminary 10-question generation evaluation, the Top-20 system earned 18 of 20 possible points. These results guided the second system version and motivated a fact-checking and evidence-filtering phase.

1 Introduction

News about major events is often reported across several languages, and different articles may contain different parts of the same story. This can make it difficult for readers to find complete and reliable information when useful sources are written in a language they do not understand. A reader may also need to compare multiple articles before finding enough evidence to answer a specific question. This project focuses on making that process easier by building a multilingual question-answering system that can process supplied news articles and return an answer in the user's preferred language.

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The system uses retrieval-augmented generation (RAG), an approach that combines external evidence retrieval with language-model generation [5]. Instead of asking a large language model to answer only from information learned during training, the retrieval component first searches the available articles for information related to the user's question. The selected evidence is then passed to a language model to create the final response. Retrieval accuracy is therefore one of the most important parts of the system. If the retriever misses important information or selects a similar but unrelated source, the generated answer may be incomplete, inaccurate, or unsupported.

Long news articles and distracting documents make this retrieval process more difficult. Some articles exceed the input length supported by an embedding model, which can cause later portions of the article to be excluded if the text is processed as one unit. The document collection may also include articles that are only partly related to the question or that describe a different part of the same event. Because these documents can contain similar names, dates, locations, and numerical details, the system may retrieve information that appears relevant but does not answer the question. This becomes especially challenging when a complete answer requires evidence from more than one article.

The prototype developed in this project is designed to accept news articles and questions in English, Spanish, or Arabic and to return an answer in the selected language. Its main goal is to improve how accurately the system identifies the information required for an answer, especially when useful evidence is spread across multiple documents or languages. Although the prototype supports three user languages, the evaluation reported in this chapter focuses on Arabic and Spanish questions. English appears as an evidence language in XRAG rather than as a separate question-language group in the selected sample. The project evaluates retrieval separately from generation so that failures can be traced to either missing evidence or incorrect use of evidence that was successfully retrieved.

To evaluate the system, I used a 50-question subset of the XRAG benchmark [6]. The subset contained 25 Arabic questions and 25 Spanish questions. Each example included a gold answer, eight candidate news articles, and article-level support labels. Two articles contained the information required for the answer, while the remaining six were topically related distractors. This structure made XRAG useful for testing whether a retriever could recover both supporting articles while avoiding similar but insufficient evidence.

I compared four retrieval configurations while developing the system: BM25 over full articles, multilingual E5 over overlapping text chunks, a hybrid BM25–E5 method, and BGE-M3 over individual sentences. BM25 performed best among the initial configurations, while the hybrid method did not improve retrieval as expected. After switching to sentence-level BGE-M3 retrieval, the system represented both supporting articles for 40 of the 50 questions within the Top-20 results and achieved an average support-article recall of 0.88. These findings guided the design of the second system version and supported moving into the next phase of the project, which focuses on filtering evidence and checking whether generated claims are supported.

The main research question guiding this project is: *How do retrieval granularity and the Top-K evidence budget affect complete-support retrieval and generated-answer quality for Arabic and Spanish XRAG questions?*

The remainder of this chapter reviews the research that guided the system, describes the XRAG data and retrieval methods, presents the retrieval and preliminary generation results, and examines the main errors observed during testing. The chapter concludes with potential applications, current limitations, and the next steps for improving fact-checking, answer reliability, and usability.

2 Related Work and Background

Building this system required studying previous work in retrieval-augmented generation, multilingual question answering, sparse and dense information retrieval, and long-context evidence selection. This research helped separate the retrieval problem from the generation problem and guided the decision to compare article-, chunk-, and sentence-level evidence units.

2.1 Retrieval-Augmented Generation

RAG combines a retrieval stage with a generation stage [5, 4]. When a user submits a question, the retriever searches an external collection and selects documents or text segments judged to be relevant. The question and selected evidence are then supplied to a language model, which produces the final answer. Unlike a language model operating without external context, a RAG system can use documents that were not included in the model's original training data.

Supplying external documents does not guarantee an accurate answer. The retriever may fail to find required evidence, or it may return topically related information that does not support the answer. The generator may also choose the wrong date, number, or event even when correct evidence appears in the context. For these reasons, this project measures retrieval performance separately from generated-answer quality.

2.2 Multilingual and Cross-Lingual Question Answering

Multilingual question answering requires a system to understand questions, retrieve evidence, and generate answers across more than one language. In cross-lingual settings, the question and supporting documents may be written in different languages. The system must therefore identify equivalent meanings across languages rather than depending only on exact word overlap.

Ranaldi et al. compared several multilingual RAG strategies, including translating questions before retrieval, retrieving directly across languages, and translating retrieved documents into a common language before generation [8]. Their work shows that translation choices and retrieval language can change the evidence available to the generator. XRAG focuses specifically on cross-lingual RAG using recent news and includes monolingual- and multilingual- retrieval conditions [6]. Its structure tests retrieval, cross-document reasoning, distractor resistance, and response-language control.

2.3 Sparse and Dense Retrieval

Information-retrieval methods can be separated broadly into sparse and dense approaches. Sparse retrieval focuses on lexical overlap between the question and document. Dense retrieval converts the question and documents into vector representations and compares their positions in an embedding space.

BM25 is a sparse retrieval method based on a probabilistic relevance framework [9]. It considers term frequency, term rarity across the collection, and document length. This makes it useful when a question and supporting article share important names, locations, dates, or event-specific wording. BM25 is also interpretable and does not require model inference or an external API.

Lexical retrieval may struggle when a question and article express the same idea using different words or languages. Multilingual E5 is a dense embedding model trained on multilingual text pairs [10]. It is intended to place semantically related questions and passages near each other in a shared embedding space. This made it a reasonable candidate for the Arabic and Spanish questions in this project.

Dense retrieval introduces a different risk: a passage may be similar to the overall topic without containing the specific evidence required for the answer. In the pilot experiments, E5 often ranked articles that were related to the event but did not provide the details represented in the gold answer. This distinction between topic similarity and answer relevance became important when interpreting the dense-retrieval results.

2.4 Hybrid Retrieval

Sparse and dense retrieval have different strengths, so hybrid methods attempt to combine their rankings. This project used reciprocal-rank fusion (RRF), a rank-combination method that assigns a fused score based on an item's position in each ranked list [3]. For a document d, the score is

$$\mathrm{RRF}(d) = \sum_{r \in R} \frac{1}{k + r(d)}, \tag{1}$$

where R is the set of ranked lists, $r(d)$ is the rank of document d in a list, and k is a constant that reduces the effect of large rank differences lower in the lists. The project used $k = 60$.

A hybrid retriever is most useful when its component systems recover different relevant items. If one retriever contributes mostly topically related distractors, fusion can move weaker items upward and push strong results downward. The pilot results in this project illustrate that combining methods does not automatically improve performance.

2.5 Long-Context and Sentence-Level Retrieval

Long documents create a challenge because the evidence needed for an answer may appear anywhere in the text. Sending full articles to the generator can also include many unrelated details. Bagheri Nezhad and Agrawal proposed a neural-symbolic framework for multilingual long-context retrieval and reasoning [1]. The retrieval component of their approach uses BGE-M3 to select focused sentence-level evidence before later reasoning stages. This work influenced the move from full-article BM25 retrieval to sentence-level retrieval in the second system version.

BGE-M3 is a multilingual embedding model designed to support multiple retrieval functions and input granularities [2]. It can represent short sentences as well as longer documents and supports multilingual and cross-lingual retrieval. In this project, BGE-M3 dense embeddings were used to compare each question with individual sentences. This shifted part of the evidence-localization task from the generator to the retriever.

2.6 XRAG

XRAG is a benchmark for cross-lingual RAG built from recent news articles [6]. It includes questions, gold answers, candidate documents, and document-level relevance annotations. The benchmark includes monolingual-retrieval settings, where non-English questions are paired with English evidence, and multilingual-retrieval settings, where candidate evidence may appear in multiple languages.

Each example used in this project contained two supporting articles and six topically related distractors. The two supports jointly provide the information needed for the complete answer. Because XRAG labels support at the article level rather than the sentence level, sentence retrieval can be evaluated only by checking the source article of each retrieved sentence. A sentence originating from a support article is therefore described in this chapter as *support-associated*; this does not prove that the sentence itself contains the exact answer evidence.

3 Data and Methodologies

This section describes the evaluation sample, data preparation, system architecture, retrieval configurations, answer-generation process, and metrics. The project was completed in stages so that each design change could be connected to a measured result.

3.1 Data Collection and Sample Preparation

The evaluation data came from XRAG. I used 50 questions: 25 Arabic and 25 Spanish. Each example contained the question, a gold answer, eight candidate news articles, support labels, and metadata describing the question and retrieval condition. Table 1 summarizes the sample.

Although the user-facing prototype supports English, Spanish, and Arabic, the evaluation sample contains Arabic and Spanish questions only. English was not used as a separate question-language group. Instead, English appears as an article language in XRAG, especially in the monolingual-retrieval setting. The language-based results in this chapter therefore compare Arabic and Spanish question performance.

Table 1 Summary of the XRAG evaluation sample.

Characteristic	Value
Total questions	50
Arabic questions	25
Spanish questions	25
Candidate articles per question	8
Supporting articles per question	2
Distractor articles per question	6
Gold answer available	Yes
Article-level support labels available	Yes

Table 2 shows one example from the selected Spanish data. The question requires a comparison across two supporting articles, while six related articles act as distractors.

Preparing the sample required normalizing the XRAG fields used by the project scripts. The article data were stored as strings, support information appeared in arrays, and metadata fields used names such as `q_type` and `a_type`. I created checks to confirm that articles were parsed correctly, support labels were interpreted consistently, and every example could be processed before generation calls were made.

Table 2 Example from the Spanish XRAG evaluation sample.

Field	Example
Question	¿Qué organización ha ayudado a más entidades, ASPCA a los animales desde 2010 o FEMA a los hogares a través del pago de US$750?
English translation	Which organization helped more entities: the ASPCA through animals helped since 2010, or FEMA through households receiving the US$750 payment?
Gold answer	FEMA helped more entities: more than one million households received the payment, compared with more than 128,000 animals helped by the ASPCA since 2010.
Candidate evidence	Eight news articles: two supporting articles and six topically related distractors.

3.2 System Architecture

The system follows a four-stage RAG pipeline: document processing, evidence retrieval, context preparation, and answer generation. The input consists of a question and a collection of news articles. The document-processing stage converts the articles into the retrieval unit required by the current method: full documents for BM25, overlapping token chunks for E5, or sentences for BGE-M3.

The retrieval stage assigns a relevance score to each unit and ranks the results. The highest-ranked units are retained according to a Top-K setting. The context-preparation stage combines the selected evidence with the question and instructions for the generator. Finally, the language model produces an answer, which is saved with the retrieval metadata for later scoring and error analysis.

I kept the experimental code separate from the runnable system code. The baseline folder contained retrieval tests and failure analyses, while the system folder contained setup checks, the BM25 RAG pipeline, the BGE-M3 RAG pipeline, output files, and review scripts. This separation made it easier to compare experimental configurations without mixing them with the current system implementation. Figure 1 summarizes the complete evaluation pipeline and shows where the retrieval configurations differ.



Fig. 1 Overview of the multilingual RAG evaluation pipeline. Candidate articles are converted into document-, chunk-, or sentence-level retrieval units. The Top-K evidence is passed to the generator, after which retrieval and answer quality are evaluated separately.

3.3 Initial Generation Conditions

Before comparing retrievers, I tested how evidence selection affected answer generation. The initial 20-question pilot included four conditions: question only, gold supporting documents only, all eight documents, and automatically retrieved BM25 documents. The purpose was not to select a final model, but to determine whether evidence quality was important enough to make retrieval the main system-development problem.

All generation experiments used **[INSERT EXACT GENERATOR MODEL AND VERSION]** with a controlled prompt instructing the model to answer from the supplied evidence and use the language of the question. The same prompt structure was used when comparing retrieval configurations.

3.4 BM25 Document Retrieval

BM25 treated each of the eight candidate articles as one document. The question and articles were tokenized, and the `rank-bm25` implementation assigned a score to each article. The articles were sorted by score, and Top-1 through Top-4 were evaluated. BM25 was chosen as the first automatic baseline because it was fast, interpretable, and well suited to questions containing exact names, locations, and dates.

The first runnable system version used BM25 Top-4. It loaded the XRAG example, ranked the eight articles, inserted the four highest-ranked articles into the prompt, generated an answer, and saved the answer and retrieval information for manual review.

3.5 Multilingual E5 Chunk Retrieval

The first E5 test attempted to represent each full article with one embedding. This was not a fair test because many articles exceeded the model's 512-token input limit, causing later text to be truncated. I therefore divided each article into 400-token chunks with an 80-token overlap. The overlap reduced the chance that a sentence or idea would be separated completely at a chunk boundary.

Each question and chunk was encoded with multilingual E5. Chunks were ranked by vector similarity and then mapped back to their source articles. This mapping was necessary because XRAG provides support labels at the article level. The E5 and hybrid comparison was conducted on a 20-question pilot rather than the complete 50-question sample, so the results should be interpreted as an early comparison rather than a final full-sample ranking.

3.6 BM25–E5 Reciprocal-Rank Fusion

The hybrid configuration combined BM25 article rankings with article-level rankings derived from the highest-scoring E5 chunks. Equation 1 was used with $k = 60$. RRF was selected because the raw BM25 and cosine-similarity scores use different scales and therefore should not be added directly. The fused rankings were evaluated on the same 20-question pilot as the chunked E5 method.

3.7 BGE-M3 Sentence Retrieval

The second system version divided all eight candidate articles into individual sentences. Each question and sentence was embedded with BGE-M3, and the sentences were ranked by dense-vector similarity. I evaluated sentence caps of Top-3, Top-5, Top-10, Top-20, and Top-50. The retrieved sentences retained their article identifiers so that the system could measure whether evidence from zero, one, or both support articles appeared in the selected context.

Sentence-level retrieval was intended to provide more focused evidence than full articles or multi-sentence chunks. However, because XRAG does not identify gold support sentences, the retrieval evaluation measures support-article representation rather than exact sentence correctness. Manual review of generated answers remains necessary to determine whether the selected sentences contain enough evidence for a correct response.

3.8 Answer Generation and Manual Review

For end-to-end testing, retrieved evidence was placed into a controlled prompt with the question. The prompt instructed the generator to use the provided evidence, avoid unsupported additions, and answer in the question language. The first system version used BM25 Top-4 documents. The second version compared BGE-M3 Top-10 and Top-20 sentence contexts on the same first 10 evaluation examples.

Generated answers were reviewed manually with a 0–2 correctness rubric. A score of 2 represented a fully correct answer, 1 represented a partially correct answer, and 0 represented an incorrect or unsupported answer.

Language correctness and qualitative error types were also recorded. Because the generation study contains only 10 questions and one current reviewer, the scores are preliminary.

3.9 Evaluation Metrics

Retrieval was evaluated separately from generation. Let S_q be the set of two support-article identifiers for question q, and let A_q^K be the set of article identifiers represented by the Top-K retrieved units. Support-article recall for one question is

$$\text{Recall@}K(q) = \frac{|S_q \cap A_q^K|}{|S_q|}. \tag{2}$$

Because each example has two supporting articles, the value is 0, 0.5, or 1.0. The reported average recall is the mean across questions.

Complete-support coverage counts questions for which both support articles are represented:

$$\text{Complete@}K = \sum_q \mathbf{1}[S_q \subseteq A_q^K]. \tag{3}$$

Any-support coverage counts questions for which at least one supporting article is represented:

$$\text{Any@}K = \sum_q \mathbf{1}[S_q \cap A_q^K \neq \emptyset]. \tag{4}$$

The pilot comparison also used mean reciprocal rank (MRR), calculated from the rank of the first relevant item. MRR is treated as a secondary metric because it measures only the first relevant result and does not show whether both supporting articles were recovered.

Generation quality was measured with the 0–2 manual rubric described in Sect. 3.8. The percentage reported in the results tables was calculated as

$$\text{ManualAnswerScore} = \frac{\text{points earned}}{2N} \times 100, \tag{5}$$

where N is the number of evaluated questions. For example, 15 points out of 20 possible points equals a manual answer score of 75%. These metrics make it possible to distinguish no support, partial support, and complete support, and then compare those retrieval outcomes with answer quality.

4 Results

The results are presented in the order the experiments were completed. This chronological organization shows how each finding changed the next system design decision.

4.1 Initial Evidence-Condition Pilot

The first 20-question pilot compared generation with no documents, only gold supporting documents, all eight documents, and BM25 Top-2 documents. Table 3 reports the project summary percentages.

Table 3 Initial 20-question generation conditions.

Evidence condition	Manual answer score
Question only	67.5%
Gold support documents only	75.0%
All eight documents	65.0%
BM25 Top-2 documents	77.5%

The model performed worse with all eight articles than with clean support-only evidence. This result indicated that additional context could introduce distractors rather than automatically improve the answer. BM25 Top-2 produced the highest score in this pilot, which motivated a more detailed study of automatic evidence retrieval.

4.2 BM25 Retrieval Results

Table 4 presents BM25 retrieval on all 50 questions. Both complete-support coverage and average support recall increased as more articles were retained.

Table 4 BM25 document-level retrieval on 50 XRAG questions.

Setting	Average recall	Complete support	Any support
Top-1	0.250	7/50	18/50
Top-2	0.540	20/50	34/50
Top-3	0.650	25/50	40/50
Top-4	0.740	32/50	42/50

BM25 Top-4 recovered both supporting articles for 32 questions and became the retrieval setting for System V1. However, 18 questions still lacked complete evidence. Increasing Top-K also meant sending more distractor documents to the generator. The BM25 result therefore demonstrated the main system trade-off: higher support recall can also increase irrelevant context.

4.3 Dense and Hybrid Retrieval Pilot

Table 5 compares BM25, chunked multilingual E5, and BM25–E5 RRF on the 20-question pilot.

Table 5 Twenty-question retrieval pilot.

Retriever	Average recall at Top-2	Complete support	Any support
BM25	0.625	10/20	15/20
Chunked multilingual E5	0.350	4/20	10/20
BM25–E5 RRF	0.500	8/20	12/20

BM25 was strongest in this pilot. It also achieved an MRR of 0.773, compared with 0.568 for chunked E5 and 0.685 for the hybrid. Chunked E5 recovered both supports for 4 of 20 questions at Top-2, while the hybrid recovered both for 8. E5 did not add unique complete-support successes at Top-2, so the fusion method had limited opportunity to improve BM25. In several cases, topically related E5 results changed the combined ranking without adding the missing supporting article. This was the most surprising early result because the hybrid was expected to combine lexical and semantic strengths, but it instead performed below BM25 alone.

The E5 experiment still provided an important methodological result. Full-article embedding was not appropriate because of truncation, and chunking corrected that weakness. The lower score therefore cannot be explained only by failing to process the article length. On this pilot, semantic similarity often reflected the correct topic without identifying the answer-bearing source.

4.4 System V1 Generation Results

System V1 connected BM25 Top-4 retrieval to controlled answer generation. On the first 10 examples, the generated answers earned 15 of 20 possible points, or 75%. All 10 answers used the required Arabic or Spanish response language.

This stage demonstrated that the complete pipeline could run from question input through retrieval, prompt creation, generation, and manual scoring. It also showed that retrieval and generation were separate failure points. Some answers were wrong because required support was absent, while other answers were wrong even when the support articles were represented. This finding motivated the shift to more focused sentence-level evidence.

4.5 BGE-M3 Sentence-Retrieval Results

Table 6 reports sentence-level BGE-M3 retrieval on the 50-question sample.

Table 6 BGE-M3 support-article coverage from sentence retrieval.

Sentence cap	Average recall	Complete support	Any support
Top-3	0.550	20/50	35/50
Top-5	0.630	23/50	40/50
Top-10	0.770	31/50	46/50
Top-20	0.880	40/50	48/50
Top-50	1.000	50/50	50/50

Top-20 represented both supporting articles for 40 of 50 questions, compared with 32 of 50 for BM25 Top-4. It also represented at least one support article for 48 questions and achieved 0.88 average recall. Top-50 reached both support articles for all 50 questions, showing that the support sources were reachable within the ranked sentence list. However, sending 50 sentences to the generator would increase context length, repetition, and distractor exposure. Top-20 was therefore selected as the best practical sentence cap tested at this stage.

The comparison should be interpreted with care because BM25 Top-4 returns four full articles, while BGE-M3 Top-20 returns 20 sentences. The two settings do not have equal token budgets. The result establishes the behavior of the tested systems, but a final conference comparison should also control the number of tokens passed to the generator.

4.6 Results by Question Language

Table 7 separates BGE-M3 Top-20 retrieval by question language. Arabic and Spanish showed the same overall pattern: Top-20 represented at least one support article for 24 of 25 questions in each group. Arabic had slightly higher average recall and complete- support coverage in this sample.

Table 7 BGE-M3 Top-20 results by question language.

Language	Questions	Average recall	Complete support	Any support
Arabic	25	0.900	21/25	24/25
Spanish	25	0.860	19/25	24/25

This difference should not be interpreted as evidence that Arabic retrieval is generally easier than Spanish retrieval. The two groups contain different questions

rather than translations of the same examples, so the result may reflect question type, retrieval condition, document wording, or example difficulty.

4.7 System V2 Generation Results

The first 10 questions were generated with BGE-M3 Top-10 and Top-20 evidence. Table 8 compares the two settings with System V1.

Table 8 Preliminary generation scores on the first 10 questions.

System	Manual answer score	Complete support	Average recall
BM25 Top-4 documents	15/20	–	–
BGE-M3 Top-10 sentences	16/20	5/10	0.700
BGE-M3 Top-20 sentences	18/20	8/10	0.900

BGE-M3 Top-20 produced the strongest preliminary generation score: 18 of 20 points, or 90%. It also represented complete support for 8 of the 10 questions, compared with 5 under Top-10. The results are consistent with the idea that recovering more complete evidence can improve answer quality. However, the sample is too small to establish a general performance difference, and the conditions were not matched by token count.

4.8 Error Analysis

The remaining errors were concentrated in several repeated patterns. First, the retriever sometimes returned evidence from similar events with different dates or death counts. Second, the correct support source could be represented while the generator selected the wrong number or detail. Third, several highly ranked sentences repeated the same information or came from the same article, limiting the diversity of the context. Fourth, some answers captured the main comparison correctly but included one incorrect numerical detail or an unsupported addition.

These errors show that support-article representation is necessary but not sufficient. A sentence from a support article may discuss the general event without containing the exact evidence needed by the question. In addition, a generator can misuse correct evidence. Therefore, improving retrieval recall alone will not produce a fully reliable system. The next version needs to filter repeated evidence, preserve coverage across source articles, and verify generated claims against the selected text.

The following two cases were selected to illustrate distinct failure modes; they are qualitative examples rather than estimates of how often each error occurs.

4.8.1 Event mismatch despite complete support

For the question "Which location suffered more deaths from Israeli attacks, the Nuseirat refugee camp or Beirut, Lebanon?", the gold answer identified Nuseirat with 22 deaths, compared with at least 11 in Beirut. BGE-M3 Top-20 represented both supporting articles, but the generated answer stated that Nuseirat had at least 274 deaths. The system selected the correct location but used a casualty figure from a different Nuseirat event. This example shows that complete support does not guarantee that the generator will select the correct event-specific detail.

4.8.2 Wrong number with incomplete support

For the question "Which report indicates a higher number of Palestinian deaths in the Israel–Hamas conflict, the first or the second set?", the gold answer selected the second set with more than 42,000 deaths. The Top-20 answer also selected the second set but reported more than 37,000 deaths. Only one support article was represented in the retrieved context. The comparison was correct, but incomplete evidence contributed to an incorrect numerical detail.

4.9 Cross-Method Comparison

All retrieval configurations ranked the same eight candidate articles for each question. What differed was the number of questions evaluated, the retrieval unit, and the Top-K cutoff. Table 9 places the results in one table so that the directly comparable experiment groups are clear.

Table 9 Retrieval results across the completed experiment groups. The first three rows are directly comparable on the 20-question pilot; the last two rows are directly comparable on the 50-question sample.

Experiment	Method	Unit	Questions	Top-K	Avg. recall	Complete	Any support
Pilot	BM25	Article	20	2	0.625	10/20	15/20
Pilot	Multilingual E5	Chunk	20	2	0.350	4/20	10/20
Pilot	BM25–E5 RRF	Article rank	20	2	0.500	8/20	12/20
Full sample	BM25	Article	50	4	0.740	32/50	42/50
Full sample	BGE-M3	Sentence	50	20	0.880	40/50	48/50

The table clarifies that the methods did not use different numbers of candidate articles: every question contained eight. The E5 and hybrid results used a smaller 20-question pilot, while BM25 Top-4 and BGE-M3 Top-20 were both evaluated on the same 50-question sample. The latter comparison shows higher support coverage for BGE-M3, although the retrieval units and token budgets are not matched.

5 Conclusion

This project developed and evaluated two working versions of a multilingual RAG pipeline for Arabic and Spanish news questions. The first version used BM25 to retrieve full articles. The second used BGE-M3 to retrieve individual sentences. The experiments showed that the choice of retrieval unit changed how often the system represented both required support sources.

BM25 was the strongest early baseline, recovering complete support for 32 of 50 questions at Top-4. Multilingual E5 and BM25–E5 fusion did not outperform BM25 in the 20-question pilot. BGE-M3 Top-20 represented complete support for 40 of 50 questions and reached 0.88 average support-article recall. In preliminary generation tests on 10 questions, the BGE-M3 Top-20 system earned 18 of 20 points, compared with 15 of 20 for the earlier BM25 system.

The results support two main conclusions. First, multilingual RAG should evaluate retrieval separately from generation because errors can occur at either stage. Second, increasing support coverage is helpful, but the evidence supplied to the generator must also be controlled. Similar events, repeated sentences, and incorrect details remained problems even when the support articles were represented.

5.1 Applications

A system of this type could help readers examine news coverage written in languages they do not speak. It could also support journalists, researchers, students, and community organizations that need to compare information across multilingual sources. The same architecture could be adapted to other document collections, including public reports, educational materials, and organizational archives.

The current work should not be treated as a replacement for professional fact-checking. Its most realistic near-term use is as an evidence- navigation tool that helps users locate relevant passages and inspect their sources. Displaying the evidence used for an answer would make the system more transparent and allow users to verify important claims.

5.2 Concerns and Limitations

The study has several limitations. The retrieval evaluation used only 50 questions from two languages, and the dense/hybrid pilot used only 20. The generation comparison used 10 questions and one manual reviewer. The language balance does not represent all XRAG languages or the full range of multilingual news conditions. The current results therefore describe this sample rather than multilingual RAG in general.

XRAG supplies article-level support labels but not gold support sentences. The BGE-M3 metric measures whether retrieved sentences came from support articles, not whether each sentence contained the exact answer evidence. In addition, BM25 and BGE-M3 were compared at different retrieval granularities and context sizes. A stronger final evaluation should compare matched token budgets and rerun all retrievers on the same full sample.

Generated news answers also create a risk of misinformation. Even when the system is grounded in supplied articles, it may combine events, select the wrong number, or add unsupported information. The system should expose its evidence, avoid presenting uncertain answers as facts, and make clear that important claims require verification.

5.3 Next Steps

The next system version will add an evidence-control stage between BGE-M3 retrieval and generation. The planned process will begin with the Top-20 sentence list, remove duplicate or nearly duplicate sentences, limit excessive concentration from one article, preserve evidence from multiple sources, and rerank the remaining items into a smaller final context. This design aims to keep the recall advantage of Top-20 while reducing repetition and distractor exposure.

The answer-generation evaluation should then be expanded to all 50 questions. The final study should record the exact model version, temperature, token limits, and prompts; use at least two reviewers when possible; report agreement between reviewers; and compare systems using the same questions and matched evidence budgets. Further work could also add claim-level fact checking that links each generated statement to a supporting sentence or marks it as unsupported.

A user-facing version should support clear article input, language selection, evidence display, and source tracing. Rather than only returning an answer, the interface should show which passages were used and allow the reader to inspect the original article context.

5.4 REU Experience

The most important lesson I learned while building this system was to slow down and verify every stage of the process. This project was different from a typical class assignment because the goal was to produce research that could be published and eventually inform a tool used by other people. That required checking not only answer accuracy, but also data parsing, retrieval behavior, code organization, output files, and the clarity of the user workflow.

I also learned that literature review is part of system building rather than a separate activity. Studying XRAG established a controlled evaluation problem, while work

on multilingual sentence retrieval provided a direction when the first approaches reached their limits. The project improved when each design decision was connected to either previous research or an observed failure in the experiments.

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